

LOCALIZED BOST–CONNES SYSTEMS V: THE FREE VARIANT AND THE FREE CRITICAL EXPONENT

RUQING CHEN

ABSTRACT. Replacing the free abelian monoid J_S of the localized Bost–Connes system by the free monoid $\mathbb{F}_S^+$ produces a system whose statistics are free-probabilistic and whose thermodynamics is governed by a new arithmetic invariant. We determine both, and conclude that the deformation is incompatible with the phase structure it was meant to probe.

The incompatibility begins at the definition: keeping the coefficient algebra $C(Y_S)$ while freeing the isometries is inconsistent, since orthogonal range projections contradict the nested ones of $C(Y_S)$, and the universal algebra is zero. The consistent free system is the Toeplitz–Cuntz algebra of $\mathbb{F}_S^+$ with the norm dynamics; it retains the norms of S and nothing else — no symmetry group G_S , no obstruction group Ξ_β . Its partition function is $Z_{\text{free}}(\beta) = (1 - L_S(\beta))^{-1}$ with $L_S(\beta) = \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta}$, so the critical point solves $L_S = 1$: a Galton–Watson branching criticality rather than an abscissa of convergence. The range projections being mutually orthogonal, every KMS_β state forces $L_S(\beta) \leq 1$, so *no KMS_β state exists below the threshold*, and above it there is exactly one — the same mechanism by which partitioning, as opposed to nested, range projections restrict the admissible temperatures in the $ax + b$ systems (Paper II of the series).

In the Fock–Gibbs state the field operators $\Phi_{\mathfrak{p}} = \mu_{\mathfrak{p}} + \mu_{\mathfrak{p}}^*$ have an exactly computable law on $[-2, 2]$: with $t = N\mathfrak{p}^{-\beta}$ the moments are $m_2 = 1 + t$, $m_4 = 2 + 3t + t^2$ and the free cumulant $\kappa_4 = -t(1 + t)$ is strictly negative, so $\Phi_{\mathfrak{p}}$ is semicircular only in the limit $\beta \rightarrow \infty$. The fields are not exactly free: the obstruction is the hard-core exclusion $e_{\mathfrak{p}}e_{\mathfrak{q}} = 0$, giving the mixed cumulant $-t_{\mathfrak{p}}t_{\mathfrak{q}}$, and asymptotic freeness holds at exactly that rate. Two natural guesses fail: $\mathcal{N}_{\mathfrak{p}} = \mu_{\mathfrak{p}}\mu_{\mathfrak{p}}^*$ is a *projection*, so its law is Bernoulli and not free Poisson; and Voiculescu’s microstates free entropy is undefined for $n \geq 2$ variables, the KMS_β state not being tracial.

The invariant $\beta_c^{\text{free}}(S)$, the root of $L_S = 1$, is arithmetically of a different kind from the classical critical exponent. It is not a tail invariant: over $\mathbb{Q}$, deleting the single prime 2 moves it from 1.399433..., the root of the prime zeta function minus one, to 1.224388...; and it obeys the universal bound $\beta_c^{\text{free}}(S) \leq \beta_c^{\text{free}}(P_{\mathbb{Q}})$. Applied to the cascade constructions of [7] the conclusion is sharp and negative: there $L_S(1) = \infty$ by design, so $\beta_c^{\text{free}}(S) > 1$, while the entire transition locus lies in $(0, 1]$. Every classical phase transition occurs at a temperature where the free system has no equilibrium state at all, and β_c^{free} can be prescribed anywhere in $(1, \beta_c^{\text{free}}(P_{\mathbb{Q}}))$ without disturbing the cascade. The two exponents carry disjoint information.

CONTENTS

1. The free system	2
1.1. The naive deformation is inconsistent	2
1.2. The Toeplitz–Cuntz system	2
2. The law of the fields	3
3. Asymptotic, not exact, freeness	4
4. The free critical exponent	5
5. The free deformation annihilates the cascade	6
6. Assessment	7
References	7

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/V-free/](#).

1. THE FREE SYSTEM

1.1. The naive deformation is inconsistent. The first idea is to keep the coefficient algebra of the classical system and to free the isometries: consider the universal C^* -algebra $\mathcal{A}^{\text{naive}}$ generated by a copy of $C(Y_S)$ and isometries μ_p , $p \in S$, subject to

$$(1.1) \quad \mu_p^* \mu_q = \delta_{pq} 1, \quad \mu_p f \mu_p^* = \alpha_p(f) \quad (f \in C(Y_S)),$$

where $\alpha_p(f) = (f \circ p^{-1}) \mathbf{1}_{pY_S}$ is the endomorphism of the classical system [6, §2]. The two halves of (1.1) pull in opposite directions: the first makes the range projections $e_p = \mu_p \mu_p^*$ orthogonal, the second identifies $e_p = \alpha_p(1)$ with the image of $\mathbf{1}_{pY_S}$, and in $C(Y_S)$ the range projections are nested, $\mathbf{1}_{pY_S} \mathbf{1}_{qY_S} = \mathbf{1}_{pqY_S} \neq 0$.

Proposition 1.1 (Incompatibility). *If $|S| \geq 2$ then $\mathcal{A}^{\text{naive}} = 0$.*

Proof. Let $p \neq q$. By (1.1), $e_p e_q = \mu_p (\mu_p^* \mu_q) \mu_q^* = 0$, while $e_p e_q$ is the image of $\mathbf{1}_{pY_S} \mathbf{1}_{qY_S} = \mathbf{1}_{pqY_S} = \alpha_p(\mathbf{1}_{qY_S})$. Hence $\mu_p \mathbf{1}_{qY_S} \mu_p^* = 0$, and multiplying by μ_p^* on the left and μ_p on the right, using $\mu_p^* \mu_p = 1$, gives $\mathbf{1}_{qY_S} = 0$ in $\mathcal{A}^{\text{naive}}$. Then $\mu_q = e_q \mu_q = \mathbf{1}_{qY_S} \mu_q = 0$, so $1 = \mu_q^* \mu_q = 0$. $\square$

The classical system has a single isometry per prime with *nested* ranges, and this is what lets $C(Y_S)$, with its symmetry group G_S and obstruction group Ξ_β , sit inside it; a free monoid of isometries has *partitioning* ranges, and cannot coexist with that coefficient algebra. The consistent free deformation therefore retains, of the arithmetic of S , only the norms.

1.2. The Toeplitz–Cuntz system.

Definition 1.2. Let $\mathbb{F}_S^+$ be the free monoid on S and $Nw = \prod_j Np_j$ for $w = p_1 \cdots p_k$. The *free system* is the Toeplitz–Cuntz algebra $\mathcal{A}_S^{\text{free}} = \mathcal{T}_S$, the universal C^* -algebra generated by isometries μ_p , $p \in S$, with

$$(1.2) \quad \mu_p^* \mu_q = \delta_{pq} 1,$$

with the dynamics $\sigma_t(\mu_w) = Nw^{it} \mu_w$. Write $e_p = \mu_p \mu_p^*$, $\mu_w = \mu_{p_1} \cdots \mu_{p_k}$, and $L_S(\beta) = \sum_{p \in S} Np^{-\beta}$.

The Fock representation on $\ell^2(\mathbb{F}_S^+)$, $\mu_p \xi_w = \xi_{pw}$, is faithful: for finite S this is Cuntz’s theorem [2], and in general it follows from the gauge-invariant uniqueness theorem, the Fock representation carrying the gauge action. The diagonal $C^*\{\mu_w \mu_w^*\}$ of $\mathcal{T}_S$ is commutative and plays the role of the coefficient algebra; its spectrum is the space of finite and infinite words, the boundary of the word tree, and *not* Y_S .

The series. The present paper is V. The foundational paper is [6]; [7] is I of the series and treats the factor type and the spectrum of transitions. Papers II (ideal structure and index), III (Ξ_β as an invariant of a measure class) and IV (metric geometry) are in preparation; nothing here depends on them, and we refer to them by numeral, without citation.

Proposition 1.3. $\sum_{w \in \mathbb{F}_S^+} Nw^{-\beta} = \sum_{k \geq 0} L_S(\beta)^k = (1 - L_S(\beta))^{-1}$, convergent precisely when $L_S(\beta) < 1$.

Proof. Words of length k are indexed by S^k and N is multiplicative, so $\sum_{|w|=k} Nw^{-\beta} = L_S(\beta)^k$. $\square$

Theorem 1.4 (The equilibrium states). *Let $|S| \geq 2$. Then L_S is continuous and strictly decreasing on its domain with $L_S(0^+) > 1$ and $L_S(\infty) = 0$, so there is a unique $\beta_c^{\text{free}}(S)$ with $L_S(\beta_c^{\text{free}}) = 1$ whenever L_S exceeds 1 somewhere. Moreover:*

- (1) every KMS_β state satisfies $L_S(\beta) \leq 1$; hence for $\beta < \beta_c^{\text{free}}(S)$ no KMS_β state exists;

(2) for every β there is at most one KMS_β state, and it is given on the dense $*$ -algebra spanned by the $\mu_w \mu_v^*$ by

$$(1.3) \quad \varphi(\mu_w \mu_v^*) = \delta_{wv} N w^{-\beta};$$

(3) for $L_S(\beta) < 1$ the Fock–Gibbs state

$$(1.4) \quad \varphi_\beta(x) = (1 - L_S(\beta)) \sum_{w \in \mathbb{F}_S^+} N w^{-\beta} \langle x \xi_w, \xi_w \rangle$$

on $\ell^2(\mathbb{F}_S^+)$ is a KMS_β state, hence the unique one;

(4) at $\beta = \beta_c^{\text{free}}(S)$ a KMS_β state exists, namely the weak- $*$ limit of $\varphi_{\beta'}$ as $\beta' \downarrow \beta_c^{\text{free}}$, and it is the unique one.

So the free system has no equilibrium state below β_c^{free} and exactly one at every $\beta \geq \beta_c^{\text{free}}$.

Proof. (1) By (1.2) the e_p are mutually orthogonal, so $\sum_{p \in F} e_p \leq 1$ for finite F ; and the KMS condition gives $\varphi(e_p) = N p^{-\beta} \varphi(\mu_p^* \mu_p) = N p^{-\beta}$. Hence $\sum_{p \in F} N p^{-\beta} \leq 1$ for every finite F , i.e. $L_S(\beta) \leq 1$, and strict monotonicity forces $\beta \geq \beta_c^{\text{free}}$.

(2) The elements $\mu_w \mu_v^*$ span a dense $*$ -subalgebra, by (1.2). For a KMS_β state φ and words w, v , the KMS condition with the entire element μ_w , $\sigma_{i\beta}(\mu_w) = N w^{-\beta} \mu_w$, gives $\varphi(\mu_w \mu_v^*) = N w^{-\beta} \varphi(\mu_v^* \mu_w)$. If neither of v, w is a prefix of the other, $\mu_v^* \mu_w = 0$ by (1.2). If $w = vu$ with $u \neq \emptyset$ then $\mu_v^* \mu_w = \mu_u$, and if $v = wu$ then $\mu_v^* \mu_w = \mu_u^*$; in both cases the value vanishes, since φ is invariant under σ_t and $\mu_u^\pm$ is an eigenvector of σ_t with eigenvalue $N u^{\pm it} \neq 1$. Hence $\varphi(\mu_w \mu_v^*) = 0$ for $w \neq v$, and $\varphi(\mu_w \mu_w^*) = N w^{-\beta} \varphi(\mu_w^* \mu_w) = N w^{-\beta}$.

(3) (1.4) is a state by Proposition 1.3, and $\langle \mu_w \mu_v^* \xi_u, \xi_u \rangle \neq 0$ forces $u = vu'$ and $wu' = u$, i.e. $w = v$, so $\varphi_\beta(\mu_w \mu_v^*) = \delta_{wv} (1 - L_S) \sum_{u'} N (wu')^{-\beta} = \delta_{wv} N w^{-\beta}$, which is (1.3); the KMS condition for the generators then follows from (1.3) by the computation of (2) read backwards.

(4) The state space is weak- $*$ compact, and a weak- $*$ limit of $\text{KMS}_{\beta'}$ states with $\beta' \rightarrow \beta$ is a KMS_β state [1, Prop. 5.3.25]; the limit exists because (1.3) converges as $\beta' \rightarrow \beta_c^{\text{free}}$. $\square$

Remark 1.5 (Branching versus convergence). Classically the partition function is the Euler product $\prod_p (1 - N p^{-\beta})^{-1}$ and the critical exponent is the abscissa of convergence of L_S . Freely, $\mathbb{F}_S^+$ has exponentially many words, $L_S(\beta)$ is the branching number of the weighted word tree, and the transition is the Galton–Watson criticality $L_S = 1$: a tree of branching number > 1 has too many words to normalize. The same orthogonality that yields $L_S(\beta) \leq 1$ produces the constraint $\beta = 1$ on the affine $(ax + b)$ system, whose Cuntz relations are finite partitions (Paper II). Nested range projections leave every temperature admissible; partitioning ones do not. And the classical phase structure does not survive at all: by Proposition 1.1 the free system has no coefficient algebra $C(Y_S)$, hence no symmetry group G_S , no obstruction group Ξ_β , and, by Theorem 1.4(2), a unique equilibrium state wherever it has one. Everything below is computed in the Fock–Gibbs state.

2. THE LAW OF THE FIELDS

Fix p , write $t = N p^{-\beta} \in (0, 1)$, $\Phi = \mu_p + \mu_p^*$, $e = e_p$.

Theorem 2.1 (Moments and cumulants). *In φ_β ,*

$$m_1 = m_3 = 0, \quad m_2 = 1 + t, \quad m_4 = 2 + 3t + t^2, \quad \kappa_4 = m_4 - 2m_2^2 = -t(1 + t) < 0,$$

and the law of Φ_p is

$$d\rho_t(x) = (1 - t) \sum_{n \geq 0} t^n U_n(x/2)^2 \frac{\sqrt{4 - x^2}}{2\pi} dx,$$

supported on $[-2, 2]$ for every β , with U_n the Chebyshev polynomials of the second kind. In particular Φ_p is not semicircular for $t > 0$.

Proof. Only gauge-invariant words survive, so odd moments vanish and $m_2 = \varphi(\mu\mu^* + \mu^*\mu) = t+1$. For m_4 , using $\mu^*\mu = 1$ and $\varphi(\mu x \mu^*) = t\varphi(x)$, the six balanced words give $\varphi(\mu\mu\mu^*\mu^*) = t^2$, $\varphi(\mu\mu^*\mu\mu^*) = \varphi(e) = t$, $\varphi(\mu\mu^*\mu^*\mu) = t$, $\varphi(\mu^*\mu\mu\mu^*) = t$, $\varphi(\mu^*\mu\mu^*\mu) = 1$, $\varphi(\mu^*\mu^*\mu\mu) = 1$, summing to $t^2 + 3t + 2$. For the density, note that $C^*(\mu_{\mathbf{p}})$ is the Toeplitz algebra, and that on the Fock space $\ell^2(\mathbb{F}_S^+) = \bigoplus_u \ell^2(\mathbf{p}^{\mathbb{N}}u)$, the sum over words u not beginning with $\mathbf{p}$, the operator Φ is the Jacobi matrix with unit off-diagonal and zero diagonal on each summand $\ell^2(\mathbb{N})$, whose spectral measure at the n -th basis vector is $U_n(x/2)^2 \sqrt{4-x^2} dx/2\pi$; grouping the terms of (1.4) by u , the law of Φ is the geometric mixture with weights $(1-t)t^n$, since the factor $\sum_u Nu^{-\beta}$ over words not beginning with $\mathbf{p}$ is $(1-t)/(1-L_S(\beta))$. The relation $\kappa_4 = m_4 - 2m_2^2$ for a centred variable is the moment-cumulant formula [3]. $\square$

Corollary 2.2 (Semicircular limit). *As $\beta \rightarrow \infty$, ρ_t converges weakly to the Wigner semicircular law of radius 2 [5]; equivalently $\kappa_2 \rightarrow 1$, all free cumulants of order ≥ 3 vanish, and the even moments converge to the Catalan numbers $1, 2, 5, 14, 42, \dots$*

Theorem 2.3 (The number operators are projections). $\mathcal{N}_{\mathbf{p}} = e_{\mathbf{p}}$ satisfies $e_{\mathbf{p}}^2 = e_{\mathbf{p}}$, so $\text{law}(\mathcal{N}_{\mathbf{p}}) = (1-t)\delta_0 + t\delta_1$ in φ_{β} . This is a Bernoulli law, not a free Poisson law: a free Poisson variable is a projection only in the degenerate cases δ_0 and δ_1 .

Proof. $\mu_{\mathbf{p}}$ is an isometry, so $\varphi(e_{\mathbf{p}}^n) = \varphi(e_{\mathbf{p}}) = t$ for every $n \geq 1$, which determines the stated law; its spectrum is $\{0, 1\}$, whereas a nondegenerate Marchenko–Pastur law has an absolutely continuous part. $\square$

3. ASYMPTOTIC, NOT EXACT, FREENESS

Theorem 3.1 (The obstruction). *For $\mathbf{p} \neq \mathbf{q}$ the fields $\Phi_{\mathbf{p}}, \Phi_{\mathbf{q}}$ are not free in φ_{β} : while $\varphi(\Phi_{\mathbf{p}}\Phi_{\mathbf{q}}) = 0$ and $\varphi(\Phi_{\mathbf{p}}\Phi_{\mathbf{q}}\Phi_{\mathbf{p}}\Phi_{\mathbf{q}}) = 0$, one has*

$$(3.1) \quad \varphi(\Phi_{\mathbf{p}}^2\Phi_{\mathbf{q}}^2) - \varphi(\Phi_{\mathbf{p}}^2)\varphi(\Phi_{\mathbf{q}}^2) = -t_{\mathbf{p}}t_{\mathbf{q}}.$$

Proof. Gauge invariance requires each letter to appear equally often with and without adjoint, so $\varphi(\Phi_{\mathbf{p}}\Phi_{\mathbf{q}}) = 0$. For $\Phi_{\mathbf{p}}^2\Phi_{\mathbf{q}}^2$ the surviving terms give $\varphi((e_{\mathbf{p}}+1)(e_{\mathbf{q}}+1)) = \varphi(e_{\mathbf{p}}e_{\mathbf{q}}) + t_{\mathbf{p}} + t_{\mathbf{q}} + 1$, and $e_{\mathbf{p}}e_{\mathbf{q}} = \mu_{\mathbf{p}}(\mu_{\mathbf{p}}^*\mu_{\mathbf{q}})\mu_{\mathbf{q}}^* = 0$ by (1.2); subtract $(1+t_{\mathbf{p}})(1+t_{\mathbf{q}})$. For the alternating moment the four balanced terms are $\mu_{\mathbf{p}\mathbf{q}}\mu_{\mathbf{q}\mathbf{p}}^*$, $\mu_{\mathbf{p}}\mu_{\mathbf{p}\mathbf{q}}^*\mu_{\mathbf{q}}\mu_{\mathbf{q}\mathbf{p}}^*$, $\mu_{\mathbf{p}}^*\mu_{\mathbf{q}}\mu_{\mathbf{p}}\mu_{\mathbf{q}}^*$ and $\mu_{\mathbf{q}\mathbf{p}}^*\mu_{\mathbf{p}\mathbf{q}}$; the second and third vanish by $\mu_{\mathbf{p}}^*\mu_{\mathbf{q}} = 0$, the fourth is $\mu_{\mathbf{q}\mathbf{p}}^*\mu_{\mathbf{p}\mathbf{q}} = 0$, and the first is $\varphi(\mu_w\mu_v^*)$ with $w \neq v$, hence 0 by (1.3). $\square$

Corollary 3.2 (Asymptotic freeness). *Let $(\mathbf{p}_k) \subset S$ with $N\mathbf{p}_k \rightarrow \infty$ and $\beta > \beta_c^{\text{free}}(S)$. Every mixed free cumulant of $\{\Phi_{\mathbf{p}_k}\}$ is $O(\prod_j t_{\mathbf{p}_{i_j}})$, hence $O(\min_j N\mathbf{p}_{i_j}^{-\beta})$, and tends to 0; the family is asymptotically free and by (3.1) the rate $t_{\mathbf{p}}t_{\mathbf{q}}$ is attained.*

Proof. By (1.3) a mixed moment $\varphi(\Phi_{\mathbf{p}_{i_1}} \cdots \Phi_{\mathbf{p}_{i_n}})$ is a sum, over the choices of μ or μ^* in each factor, of terms $\varphi(\mu_w\mu_v^*)$ after reduction by (1.2); a term survives only if it reduces to $\mu_w\mu_w^*$, and then equals $Nw^{-\beta} = \prod t$ over the letters of w with multiplicity. Every letter distinct from those reducing to the identity contributes a factor $t_{\mathbf{p}}$, so the moment differs from the corresponding moment of free variables with the same one-variable laws by terms each carrying at least one factor $t_{\mathbf{p}}$ per additional distinct letter. Möbius inversion on $\text{NC}(n)$ [3], a finite linear combination of products of moments, preserves this order of magnitude, and each $t < 1$. $\square$

Remark 3.3 (Hard-core exclusion). Free variables would satisfy $\varphi(e_{\mathbf{p}}e_{\mathbf{q}}) = t_{\mathbf{p}}t_{\mathbf{q}}$; the true value is 0, because no vector lies simultaneously in the ranges of $\mu_{\mathbf{p}}$ and $\mu_{\mathbf{q}}$. The discrepancy is exactly $-t_{\mathbf{p}}t_{\mathbf{q}}$, and as the norms grow the ranges become φ -small and the exclusion invisible. This is the free counterpart of the classical asymptotic independence of distinct primes for the geometric measures π_{β} .

Theorem 3.4 (Free entropy is unavailable). φ_β is not tracial: $\varphi_\beta(\mu_p \mu_p^*) = t_p < 1 = \varphi_\beta(\mu_p^* \mu_p)$. Hence Voiculescu's microstates free entropy $\chi(\Phi_{p_1}, \dots, \Phi_{p_n})$ [4] and the free Fisher information are undefined for $n \geq 2$, both requiring a tracial W^* -probability space. For $n = 1$ the entropy is defined, the distribution of a single self-adjoint element being a measure, and equals $\chi(\Phi_p) = \iint \log|x-y| d\rho_t(x) d\rho_t(y) + \frac{3}{4} + \frac{1}{2} \log 2\pi$; it is finite and continuous in t , and since $t_p < L_S(\beta_c^{\text{free}}) = 1$ strictly, it remains finite as $\beta \downarrow \beta_c^{\text{free}}$. There is no logarithmic singularity of the free entropy at the critical point.

Proof. The displayed inequality is the KMS condition. Both invariants are defined for n -tuples in a tracial von Neumann algebra, and no such structure is available. For $n = 1$, ρ_t is supported in $[-2, 2]$ with bounded density for t bounded away from 1, so the logarithmic energy is finite and continuous. $\square$

Remark 3.5 (Two things called free). The quantity $-\log(1 - L_S(\beta)) = \log Z_{\text{free}}(\beta)$ does diverge as $\beta \downarrow \beta_c^{\text{free}}$, by Proposition 1.3. That is the free *energy*, a thermodynamic quantity attached to the whole algebra, not Voiculescu's free *entropy*, an invariant of an n -tuple of noncommutative random variables. The coincidence of names is unfortunate; an asymptotic of the form $\chi_\beta \sim \frac{n}{2} \log(1 - L_S(\beta))$ conflates them, its right side being the free energy and its left side undefined for $n \geq 2$ and bounded for $n = 1$.

4. THE FREE CRITICAL EXPONENT

Definition 4.1. Let $\beta_c^{\text{cl}}(S) = \inf\{\beta : L_S(\beta) < \infty\}$ be the abscissa of convergence, the classical critical exponent of [6], and $\lambda_S = \lim_{\beta \downarrow \beta_c^{\text{cl}}(S)} L_S(\beta) \in (0, \infty]$. When $\lambda_S > 1$ let $\beta_c^{\text{free}}(S)$ be the root of $L_S = 1$; when $\lambda_S \leq 1$ set $\beta_c^{\text{free}}(S) = \beta_c^{\text{cl}}(S)$.

Theorem 4.2 (Trichotomy). *For infinite S exactly one of the following holds.*

- (S) $\lambda_S > 1$: there is a unique $\beta_c^{\text{free}}(S) \in (\beta_c^{\text{cl}}(S), \infty)$, the admissible set is $[\beta_c^{\text{free}}, \infty)$, and $Z_{\text{free}} \rightarrow \infty$ as $\beta \downarrow \beta_c^{\text{free}}$.
- (C) $\lambda_S = 1$: $\beta_c^{\text{free}} = \beta_c^{\text{cl}}$, the admissible set is $(\beta_c^{\text{cl}}, \infty)$, and $Z_{\text{free}} \rightarrow \infty$ at β_c^{cl} .
- (B) $\lambda_S < 1$: $\beta_c^{\text{free}} = \beta_c^{\text{cl}}$, the admissible set is $(\beta_c^{\text{cl}}, \infty)$, and Z_{free} is bounded there, with supremum $(1 - \lambda_S)^{-1} < \infty$: the free phase is closed.

For every $\alpha \in (0, 1)$ and every $\lambda \in (0, \infty]$ there is $S \subseteq P_{\mathbb{Q}}$ with $\beta_c^{\text{cl}}(S) = \alpha$ and $\lambda_S = \lambda$, so all three occur.

Proof. Monotonicity and continuity of a Dirichlet series on its half-plane of convergence give the trichotomy, using Theorem 1.4(1) for the admissible sets. For the realization, fix α . A set T with $|T \cap [1, x]| \asymp x^\alpha$, obtained greedily as in [7, Lem. 5.4], has $\beta_c^{\text{cl}}(T) = \alpha$ and, by Abel summation, $L_T(\alpha) = \infty$, giving $\lambda = \infty$. For $\lambda < \infty$ take instead $|T \cap [1, x]| \asymp x^\alpha / \log^2 x$; then $\beta_c^{\text{cl}}(T) = \alpha$ still while the m -th element satisfies $\ell_m \asymp (m \log^2 m)^{1/\alpha}$ and $\ell_m^{-\alpha} \asymp (m \log^2 m)^{-1}$, so $L_T(\alpha) < \infty$. Discarding an initial segment of T lowers $L_T(\alpha)$ below λ without changing β_c^{cl} ; then adjoin primes $p_1 < p_2 < \dots$ outside T chosen greedily, p_k so large that $p_k^{-\alpha}$ is at most half of the remaining deficit $\lambda - L(\alpha)$ of the set built so far. The deficit tends to 0, the adjoined set has counting function $O(\log x)$, hence does not change β_c^{cl} , and the resulting S has $L_S(\alpha) = \lambda$ exactly. $\square$

Proposition 4.3 (Not a tail invariant). $\beta_c^{\text{cl}}(S)$ is unchanged by adding or deleting finitely many primes and depends only on the counting function, through $\beta_c^{\text{cl}}(S) = \limsup_m \log m / \log \mathbf{N} p_m$. By contrast β_c^{free} is monotone in S and strictly changed by deleting a single prime in regime (S); two sets with the same counting asymptotics can have different β_c^{free} . Moreover, over $\mathbb{Q}$,

$$\beta_c^{\text{free}}(S) \leq \beta_c^{\text{free}}(P_{\mathbb{Q}}) = 1.399433335\dots,$$

the root of the prime zeta function minus one, with equality only for $S = P_{\mathbb{Q}}$.

Proof. The first is the standard formula for the convergence exponent. The rest follows from strict monotonicity of L_S in S and in β . $\square$

Remark 4.4 (Computed values). Computed from the primes below $x = 2 \cdot 10^7$ with an analytic tail correction (code in `code/verify_series.py`): substituting $t = e^u$ gives $\sum_{p>x} \mathbf{N} p^{-\beta} \approx \int_{\log x}^{\infty} e^{(1-\beta)u} u^{-k} du$, the exponential integral $E_1((\beta - 1) \log x)$ for $k = 1$ and, for the Hardy–Littlewood densities $2C_2/\log^2 t$ with $k = 2$, a quantity finite at $\beta = 1$ with value $2C_2/\log x = 0.0785$:

S	$L_S(1^+)$	$\beta_c^{\text{free}}(S)$
all primes	∞	1.399433
all primes except 2	∞	1.224388
all primes except 2, 3	∞	1.154890
$\{p > 10^6\}$	∞	1.019162
Sophie Germain	1.535	1.2114
twin primes (lower)	1.059	1.0089

The first four rows are correct to the digits shown; the last two depend on the tail model in their third decimal. Removing the single prime 2 moves β_c^{free} by 0.175, quantifying Proposition 4.3; nothing comparable can happen to β_c^{cl} , which is 1 for the first four rows. The twin row should be read with care: the bare sum over the computed range is 0.9805 and the Hardy–Littlewood tail contributes 0.0785, so the set is supercritical but only just, and the conclusion rests on the tail estimate rather than on a computation (Brun’s constant $B_2 = 1.9022 \dots$ gives the same picture: $\sum_p \text{twin } p^{-1} = \frac{1}{2}(B_2 + \sum 2/(p(p+2))) \approx 1.06$). Whether $\sum_p \text{twin } p^{-1}$ exceeds 1 is an arithmetic question we do not settle; if it does not, the twin set is an instance of regime (B).

5. THE FREE DEFORMATION ANNIHILATES THE CASCADE

Theorem 5.1 (Annihilation). *Let S be any of the cascade or Cantor sets of [7, Thm. 5.1, Cor. 6.2]. Then $\beta_c^{\text{cl}}(S) = 1$ and $\lambda_S = \infty$, so $\beta_c^{\text{free}}(S) > 1$ strictly; the transition locus of the classical system lies in $(0, 1]$; and by Theorem 1.4(1) the free system has no KMS_β state for $\beta < \beta_c^{\text{free}}(S)$. Consequently every classical phase transition occurs at a temperature at which the free system has no equilibrium state; on the whole admissible range of the free system the classical one has $\zeta_{K,S}(\beta) < \infty$ and $\Xi_\beta = \widehat{G}_S$, i.e. is in its Gibbs regime with all its phase transitions behind it; and the free system has a single transition, at $\beta_c^{\text{free}}(S)$, from no equilibrium state to exactly one.*

Proof. The abundant base of [7, Lem. 5.4(1)] has $\sum_{\ell \in P_0} \ell^{-1} = \infty$, so $L_S(1) = \infty$, $\beta_c^{\text{cl}}(S) = 1$ and $\lambda_S = \infty$; Theorem 4.2(S) gives $\beta_c^{\text{free}}(S) > 1$. By [7, Thm. 5.1(3)] the thresholds lie in $(0, 1)$ and $\Xi_\beta = \widehat{G}_S$ for $\beta > 1$. $\square$

Theorem 5.2 (The two exponents are independent). *Let S be a cascade set and $\gamma \in (1, \beta_c^{\text{free}}(P_{\mathbb{Q}}))$. There is S' differing from S by finitely many primes — hence with the same chain $\{\Xi_\beta\}$ and the same transition locus — with $\beta_c^{\text{free}}(S')$ within any prescribed ε of γ . The interval is optimal: $\beta_c^{\text{free}}(S') > 1$ always by Theorem 5.1 and $\beta_c^{\text{free}}(S') < \beta_c^{\text{free}}(P_{\mathbb{Q}})$ by Proposition 4.3.*

Proof. Adding or deleting finitely many primes leaves Ξ_β unchanged, a finite set contributing a finite amount to every series in (1.2) and the computation of Ξ_β in [7, Thm. 5.1] being stated modulo finite exceptional sets. Deleting all primes of S below z gives $L_{S'}(\beta) \leq \sum_{\ell > z} \ell^{-\beta} \rightarrow 0$ for fixed $\beta > 1$ as $z \rightarrow \infty$, so $\beta_c^{\text{free}}(S')$ can be pushed arbitrarily close to 1; adjoining all primes below N gives $L_{S'} \geq \sum_{p \leq N} p^{-\beta}$, whose root increases to $\beta_c^{\text{free}}(P_{\mathbb{Q}})$. Adding one prime at a time moves the root by a controlled amount, giving every intermediate value up to ε . $\square$

Remark 5.3 (Numerical illustration). Restricting to primes above z gives the upper bounds 1.056773 at $z = 10^2$, 1.028710 at $z = 10^4$, 1.019162 at $z = 10^6$ and 1.014372 at $z = 10^8$ (the last from the tail integral alone), tending to 1; adjoining all primes below N gives roots 1.147338 at $N = 10$, 1.356444 at $N = 10^2$, 1.388390 at $N = 10^3$, 1.396110 at $N = 10^4$, 1.399069 at $N = 10^6$ and 1.399307 at $N = 10^7$, increasing to $\beta_c^{\text{free}}(P_{\mathbb{Q}}) = 1.399433$ but never attaining it.

Remark 5.4 (What the deformation is good for). The free variant was proposed as a route to importing free probability into the arithmetic phase structure. Theorem 5.1 shows the two are incompatible in a precise sense: the phenomena classified by Ξ_β live at high temperature, where the partition function diverges, and it is exactly that region which the free monoid renders inaccessible, because exponential word growth prevents normalization once the branching number exceeds 1. The classical system is interesting at high temperature and trivial at low; the free system exists only at low temperature and is trivial there. Theorem 5.2 confirms that the two exponents carry disjoint information rather than being two views of one structure.

What the deformation does produce is β_c^{free} itself, a well-defined arithmetic invariant of a set of primes: computable, universally bounded over $\mathbb{Q}$, and — unlike every exponent in this series — sensitive to individual norms rather than only to a counting function.

6. ASSESSMENT

Statement	Status	Reference
$C(Y_S)$ survives the deformation	false: the universal algebra is 0	Prop. 1.1
$Z_{\text{free}} = (1 - L_S)^{-1}$, $L_S(\beta_c^{\text{free}}) = 1$	true	Prop. 1.3, Thm. 1.4
two regimes across β_c^{free}	false: empty below	Thm. 1.4(1)
KMS $_\beta$ state unique for $\beta \geq \beta_c^{\text{free}}$	true	Thm. 1.4(2)
Φ_p semicircular	only as $\beta \rightarrow \infty$	Thm. 2.1, Cor. 2.2
$\mathcal{N}_p$ free Poisson	false: a projection	Thm. 2.3
$\{\Phi_p\}$ exactly free	false	Thm. 3.1
$\{\Phi_p\}$ asymptotically free	true	Cor. 3.2
$\chi(\Phi_{p_1}, \dots, \Phi_{p_n})$, $n \geq 2$	undefined	Thm. 3.4
β_c^{free} a tail invariant	false	Prop. 4.3
cascade survives the deformation	false	Thm. 5.1

Three questions remain. First, whether some intermediate deformation — a coefficient algebra adapted to the word tree, carrying an action of a quotient of G_S — can retain part of the classical phase structure while keeping partitioning ranges; Proposition 1.1 rules out the obvious candidate, not every candidate. Second, φ_β admits no trace, but the associated von Neumann algebra has a canonical trace on its continuous core; whether a free entropy computed there carries arithmetic information is open, and by the pattern of Paper III we expect not. Third, $\beta_c^{\text{free}}(S)$ is defined for any set of primes and we have not found an interpretation of it outside this system — for instance as a threshold for a branching process on the multiplicative semigroup generated by S — nor decided the regime of the twin primes, which by Remark 4.4 reduces to comparing a Brun-type constant with 1.

REFERENCES

- [1] O. Bratteli, D. W. Robinson, *Operator Algebras and Quantum Statistical Mechanics 2*, 2nd ed., Springer, 1997.
- [2] J. Cuntz, *Simple C^* -algebras generated by isometries*, Comm. Math. Phys. 57 (1977), 173–185.
- [3] A. Nica, R. Speicher, *Lectures on the Combinatorics of Free Probability*, LMS Lecture Notes 335, CUP, 2006.
- [4] D. V. Voiculescu, *The analogues of entropy and of Fisher's information measure in free probability theory II*, Invent. Math. 118 (1994), 411–440.
- [5] D. V. Voiculescu, K. Dykema, A. Nica, *Free Random Variables*, CRM Monogr. Ser. 1, AMS, 1992.

- [6] R. Chen, *KMS state uniqueness and phase transitions in localized Bost–Connes systems*, preprint, 2026, <https://doi.org/10.5281/zenodo.22152101>.
- [7] R. Chen, *Localized Bost–Connes systems I: factor type and the spectrum of transitions*, preprint, 2026, DOI 10.5281/zenodo.22160827.

GUT GEOSERVICE INC., MONTREAL, CANADA
Email address: ruqing@hotmail.com